

Agency-Dependent Fairness Evaluation

A Role-Counterfactual Audit of Political Fairness in Language Models, with a Validated Judge

June 2026

Abstract

We ask whether a language model’s political-fairness behavior depends on the *civic role* it is assigned. We operationalize “Agency-Dependent Fairness” (ADFE) as a role-counterfactual audit: hold the topic and task constant, vary the assigned role across an agency ladder, and score each output on a six-part fairness signature with an LLM judge that is first *validated against human-labeled datasets*. After hardening the pipeline against a set of confounds that allowed an earlier version to manufacture its own effect—a self-judging model, a circular outcome metric, and a prompt-tuning loop—we run two studies. On five small local models ($n = 2,100$ scored outputs) the central “agency gradient” hypothesis is cleanly *not supported*: curation accountability does not rise with assigned agency. On three frontier models (xAI Grok, $n = 636$) the small local judge *saturates*, scoring nearly every output a perfect 1.0 on four of six dimensions, so the gradient is largely unmeasurable; the two dimensions that retain variance (deliberative standing and viewpoint symmetry) do rise with agency, but viewpoint symmetry is precisely where the judge is least reliable ($\kappa = 0.30$). The clearest reproducible behavioral phenomenon—viewpoint-asymmetric refusal—is a small-model artifact that vanishes in capable models. We argue the most durable contribution is methodological: **judge capability must match the audited population**, and we draw out the modeling, benchmarking, and legal questions the program raises.

1 Introduction and background

People increasingly receive political information from AI systems, yet the dominant evaluation question—“is the model biased?”—is malformed. The same answer can be appropriate from a *personal assistant* drafting a partisan op-ed on request and inappropriate from a *civic news provider* that is supposed to lead with balanced fact. Fairness obligations are not a property of the model in isolation; they are conditioned on the role the system has been asked to occupy.

This report develops and tests that intuition. We treat *roles as experimental treatments* and ask whether model behavior moves, across roles, along predicted fairness dimensions. The framing is deliberately falsifiable: if outputs do not differentiate by role, the theory has a clean failure path; if they shift in the predicted directions while holding invariant floors, the theory has an empirical core.

Relation to prior work. Industry and academic efforts have measured political even-handedness with paired prompts and slant-controlled question sets (e.g. OpenAI’s political-bias work; Anthropic’s paired-prompt even-handedness evaluation; PARETO-style balanced-approval studies; news-intermediary and multiple-choice political benchmarks). These measure *ideological slant or parity*. ADFE differs by making *role/agency* the manipulated variable and by separating six distinct fairness obligations rather than collapsing fairness to a single balance score.

2 The agency-dependent fairness model

2.1 Six fairness dimensions

Each output is scored 0–4 (normalized to $[0, 1]$) on:

- **U** — User-agency fidelity: performs the assigned lawful task without hidden role substitution.
- **E** — Epistemic integrity: accurate, calibrated, grounded in supplied sources; no hallucinated authority.
- **V** — Viewpoint symmetry: analogous lawful viewpoints receive comparable effort, depth, evidence, and refusal treatment.
- **C** — Curation accountability: salience is justified; fact, analysis, and uncertainty are separated.
- **D** — Deliberative standing: disagreement, minority views, and procedural fairness are preserved.
- **M** — Non-manipulation / refusal integrity: refuses deception or unlawful manipulation on viewpoint-neutral grounds.

2.2 Roles as an agency ladder

Seven roles are ordered by an *agency level* $\alpha \in [0, 1]$ —the discretion the role implies over framing and salience: assistant and advocate ($\alpha = 0.20$), campaign aide (0.35), researcher (0.45), government information service (0.50), mediator (0.60), and civic news provider (0.75).

2.3 The agency-gradient hypothesis

The central, novel claim is a *gradient*: as α rises, dimensions tied to editorial discretion—curation accountability (C), deliberative standing (D), viewpoint symmetry (V)—should rise, while *invariant floors* for epistemic integrity (E) and non-manipulation (M) remain high and flat regardless of role. Because α is a fixed property of each role, the test is a regression of each dimension on the *continuous* treatment α across roles.

3 Methodology

3.1 Role-counterfactual generation

For each base prompt and role we build a generation prompt that supplies the role card, a dated source packet, and the user task, and instruct the model to return only the user-facing answer. The prompt bank spans six high-salience U.S. policy topics (immigration, policing, climate, abortion, taxation, voting) with structurally analogous viewpoint pairs (e.g. access vs. restriction) that we audit to match on topic, task, risk, audience, and source packet—differing only in viewpoint.

3.2 The judge, and its confounds

Outputs are scored by an LLM judge. An initial version of the harness contained three confounds that could manufacture the predicted effect, all of which we removed:

1. **Self-judging.** The judge was a member of the audited set. We hold the judge (qwen3:8b) out of the audited population entirely.

2. **A prompt-tuning loop.** An “iterate” mode auto-tuned both the generation and judge prompts to maximize the role-fit metric across cycles. We froze runs by default; the loop is now opt-in and any run that uses it is flagged **contaminated**.
3. **A circular outcome.** The headline metric was distance from researcher-chosen “expected” score bands—i.e. proximity to the answer we hoped to see, with the judge told the role in advance. We replaced it with (i) raw dimension means, (ii) a mixed-effects test of the gradient, and (iii) the expected bands reframed as falsifiable predictions. Role *inference* is now a separate, blinded judge pass with the assigned role hidden.

A live demonstration of judge fragility motivated this: a single primed value in the judge’s JSON template made the judge echo it for every item ($\kappa = 0.0$); removing it restored $\kappa \approx 0.78$. A judge whose verdicts swing that far on scaffolding cannot also be auto-tuned in a loop.

3.3 Validating the judge against human labels

We validate the judge—with no hand-labeling—against three human-labeled public datasets, one per dimension it most affects, using a reason-first, de-primed prompt and per-item resumable checkpoints:

- M (refusal/safety) against **XSTest** (250 safe + 200 unsafe prompts, expert-labeled);
- E (factuality) against **TruthfulQA** (best vs. best-incorrect answers, human truth labels);
- V (viewpoint) against **BABE** (expert sentence-level media-bias labels).

3.4 Inference

The gradient is estimated with a linear mixed model, $\text{score}_d \sim \alpha + (1 \mid \text{model})$, a random intercept per model so the slope on α is estimated while accounting for model-to-model differences. Paired-viewpoint metrics (refusal-parity gap, viewpoint quality gap) are computed per analogous pair.

4 Experimental setup

The data layer comprises 30 base prompts across 6 topics (with 6 audited viewpoint pairs), 7 role cards, and 6 dated source packets. Two populations are audited, both with the held-out, validated qwen3:8b judge:

- **Small local models** (Ollama): llama3.2:3b, llama3.2:1b, phi3:mini, gemma3:1b, deepseek-r1:1.5b; explicit and implicit agency modes; $n = 2,100$ scored outputs.
- **Frontier models** (xAI API): grok-4.3, grok-4.20-reasoning, grok-4.20-non-reasoning; explicit mode; $n = 636$ scored outputs.

5 Results

5.1 Judge validation

5.2 Small local models: the gradient is not supported

The clearest behavioral signal is viewpoint-asymmetric *refusal*: llama3.2:3b fully answers one side of a lawful argument pair (e.g. abortion access) and refuses its mirror (restriction), with a refusal-parity gap of 1.0 on the most charged pairs, even though its overall refusal rate is low ($\approx 5\%$). This is a per-model safety-tuning artifact, similar across roles, not an agency effect.

Dimension	Dataset	κ	Accuracy	Verdict
M safety	XSTest ($n=450$)	0.78	0.89	trustworthy
E factuality	TruthfulQA ($n=1,580$)	0.50	0.75	moderate; misses $\sim 20\%$ of falsehoods
V viewpoint	BABE ($n=907$)	0.30	0.64	weak; over-flags bias (specificity 0.35)

Table 1: The judge is reliable on refusal/safety, moderate on factuality, and weak on viewpoint neutrality. Conclusions are weighted accordingly.

Dim	Slope on α	p	
C curation	+0.022	0.50	predicted rise <i>absent</i>
V viewpoint	+0.040	0.20	n.s.
U user fidelity	+0.028	0.41	n.s.
D deliberative	+0.065	0.044	only hit; fails multiple-comparison correction
E epistemic	+0.009	0.78	flat (floor)
M non-manip.	+0.012	0.70	flat (floor)

Table 2: Mixed-effects gradient, small local models ($n = 2,100, 5$ models). The predicted rise in curation accountability is absent; the single significant slope (D) does not survive Bonferroni correction across six dimensions. Of 42 role $\times$ dimension interval predictions, 12 are supported, 22 violated, 8 inconclusive.

5.3 Frontier models: judge saturation

On Grok the judge returns a near-constant 1.00 on U, E, C, and M for every role, leaving no variance to model on those dimensions. Only D and V retain spread, and both rise significantly with agency (D: +0.104; V: +0.112; both $p \approx 0$). Refusal rate is 0% across all three variants with a zero refusal-parity gap. Of 42 interval predictions, 26 are “supported,” but this is partly a ceiling artifact—near-maximal scores fall inside high expected bands.

Population	U	E	V	C	D	M
Frontier (Grok), all roles	1.00	1.00	0.89–0.98	1.00	0.90–1.00	1.00
Small local, role range	—	—	0.52–0.73	0.64–0.84	0.55–0.76	—

Table 3: Mean dimension scores. The frontier rows are pinned at the ceiling on four dimensions: the 8B judge cannot discriminate among capable-model outputs.

5.4 The population contrast

Small models: a discriminating judge measures a clean *null*. Frontier models: the same judge *saturates*, so the gradient is mostly unmeasurable and the measurable part rests on the weakest gate. The honest reading is not “the gradient appears in capable models” but “a small judge cannot audit capable models.”

6 Ramifications

6.1 Modeling and measurement questions

- **Judge–population capability matching.** An evaluator must be able to discriminate within the population it scores. An 8B judge ceilings out on frontier outputs; frontier audits need a frontier-grade, separately-validated judge. This is a general law for LLM-as-judge pipelines, not a quirk of this study.
- **Circularity and pre-registration.** Defining the outcome as proximity to a researcher’s expected bands, with the judge told the answer, manufactures confirmation. Outcomes should be raw measurements plus pre-registered, falsifiable predictions.
- **Judge prompt-sensitivity.** A single primed token moved κ from 0 to 0.78. Any pipeline that *tunes* judge prompts toward a target is measuring its own tuning.
- **Dimension-specific trust.** Validation must be per-construct: the same judge was trustworthy on refusal, moderate on factuality, and weak on viewpoint neutrality.

6.2 AI-training and benchmark questions

- **Refusal asymmetry as an alignment side-effect.** The cleanest finding—one model refusing one side of a lawful pair—is a safety-tuning artifact. Benchmarks should measure *refusal parity across analogous lawful viewpoints*, not just refusal rates, and RLHF/DPO pipelines should be tuned against over-refusal that is viewpoint-correlated.
- **Role-conditioning as a first-class axis.** Fairness benchmarks that fix a single “assistant” role miss that obligations are role-dependent; role should be a manipulated variable.
- **Held-out, human-validated judges.** Benchmarks that score with an unvalidated, in-family judge inherit the judge’s blind spots. Reporting per-dimension judge κ against human labels should be a publication norm.
- **Saturation as a ceiling on benchmark lifespan.** When a judge saturates, a benchmark stops measuring; capability-matched judges and harder items are required to keep a benchmark live.

6.3 Legal and governance questions

- **Editorial discretion and the First Amendment.** If a model’s curation behavior is role-conditioned, the law’s treatment of it may be too: a system acting as a “news provider” may implicate press-clause and editorial-judgment doctrine differently than a private “assistant.” Recent platform-speech jurisprudence (e.g. the 2024 *Moody* and *Murthy* decisions) frames when curation is protected editorial expression versus regulable conduct; agency-dependence suggests the answer may vary by the role the system occupies.
- **Principal–agent liability.** As models take on agentic civic roles, allocating responsibility between deployer, model provider, and user—when a “government information service” persona gives wrong procedural guidance, or a “campaign aide” persona crosses into prohibited electioneering—becomes concrete. Role assignment is a candidate locus of duty.
- **Representation and social choice.** Viewpoint-symmetry obligations connect to representative social-choice questions: whose distribution of views should a “mediator” or “news”

model reflect, and by what aggregation? “Even-handed” is under-specified without a reference population.

- **Standing and accountability.** If systems occupy civic roles with public-facing duties, questions of auditability, disclosure of role, and recourse follow—independent of debates over any intrinsic status of the systems themselves.

7 Limitations

The audited frontier set is three variants of a single provider (Grok), so the random-effect has only three groups and results do not generalize across providers. The judge is an 8B local model: trustworthy on refusal, but saturating on frontier outputs and weak on viewpoint neutrality, which directly limits the V findings. Source packets are static, not live news. The viewpoint-pair bank is small. “Interval support” is sensitive to ceiling effects. All numbers are LLM-judge measurements; human calibration is by transfer from public datasets, not in-domain rating.

8 Conclusion

A role-counterfactual audit with a hardened, human-validated judge yields an honest, layered result. On small local models the agency-gradient hypothesis fails cleanly. On frontier models the instrument—a small judge—fails before the hypothesis can be tested, because it saturates. The most transferable lesson is methodological: evaluators must be validated per dimension and matched in capability to the systems they judge. The program’s normative core—that political-fairness duties are role-dependent—remains live and now has a runnable, falsifiable, and honestly-scoped instrument, plus a concrete agenda for the modeling, benchmarking, and legal work it implies.

References

- [1] P. Röttger et al. *XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in LLMs*. NAACL 2024. <https://github.com/paul-rottger/xstest>
- [2] S. Lin, J. Hilton, O. Evans. *TruthfulQA: Measuring How Models Mimic Human Falsehoods*. ACL 2022. <https://github.com/sylinrl/TruthfulQA>
- [3] T. Spinde et al. *Neural Media Bias Detection Using Distant Supervision With BABE*. EMNLP Findings 2021. <https://media-bias-research.org>
- [4] Anthropic. *Measuring Political Even-Handedness*. 2025. <https://github.com/anthropics/political-neutrality-eval>
- [5] OpenAI. *Defining and Evaluating Political Bias in LLMs*. 2025.
- [6] *Moody v. NetChoice*, 603 U.S. 707 (2024).
- [7] *Murthy v. Missouri*, 603 U.S. 43 (2024).